CHAPTER 1

Introduction to Categorical Data

Categorical Data Analysis

Topics Covered:

- Part A — Nature and Types of Categorical Data
- Part B — Frequency Distributions and Graphical Representation
- Part C — Cross-Classification, Binomial and Multinomial Distributions

PART A: Nature and Types of Categorical Data

1.1 Introduction to Data

Data are facts, observations, or measurements collected for the purpose of analysis and decision-making. In statistics, data are broadly classified into two categories:

- Numerical (Quantitative) Data
- Categorical (Qualitative) Data

Numerical data consist of values that can be measured or counted. Examples include height, weight, age, income, and temperature. Numerical data are further divided into:

- Discrete Data — countable values (e.g., number of students in a class)
- Continuous Data — measurable values that can take any value in a range (e.g., height, weight)

1.2 Categorical Data

Categorical data consist of categories, labels, or groups used to classify observations. The values do not represent numerical measurements; instead, they identify the group to which each observation belongs.

Definition
Categorical data are data that can be divided into distinct groups or categories based on characteristics or attributes.

Examples of Categorical Data:

Variable	Categories
Gender	Male, Female
Blood Group	A, B, AB, O
Marital Status	Married, Unmarried
Education Level	UG, PG, PhD
Mode of Transport	Bus, Train, Car

1.3 Nature of Categorical Data

Categorical data possess the following key characteristics:

1. Non-Numerical Nature

Values are represented by names, labels, or categories rather than numbers. For example: Religion, Nationality, Department.

2. Classification Property

Observations are grouped into classes based on common characteristics. For example, students may be classified according to Stream, Gender, or Grade.

3. Mutually Exclusive Categories

An observation can belong to only one category at a time. A student cannot belong simultaneously to both 'Male' and 'Female' categories.

4. Exhaustive Categories

All possible categories should be included in the classification. For example, the blood group categories A, B, AB, and O cover all possibilities.

5. Frequency-Based Analysis

Analysis is generally based on counts, frequencies, percentages, and proportions, rather than arithmetic operations on the category labels themselves.

6. No Arithmetic Meaning

Mathematical operations such as addition or averaging are usually not meaningful for categorical data. For example, computing the 'average' of blood groups A, B, AB, and O has no statistical meaning.

1.4 Types of Categorical Data

Categorical data are mainly classified into two types: Nominal Data and Ordinal Data.

1.4.1 Nominal Data

Definition
Data classified only by names or labels without any natural order or ranking are called nominal data.

Characteristics of Nominal Data:

- Categories are distinct and separate
- No ranking or order exists among categories
- Only classification is possible; arithmetic is not meaningful

Examples:

Variable	Categories
Blood Group	A, B, AB, O
Religion	Hindu, Muslim, Christian
Eye Color	Black, Brown, Blue
Nationality	Indian, American, Japanese

Important Point
In nominal data, 'A' blood group is not greater or smaller than 'B'. The categories are simply labels with no inherent order.

1.4.2 Ordinal Data

Definition
Data classified into categories that possess a meaningful order or ranking are called ordinal data.

Characteristics of Ordinal Data:

- Categories have a meaningful ranking or order

- Relative position is meaningful (e.g., 'Good' is better than 'Average')
- The exact difference between ranks may not be equal or quantifiable

Examples:

Variable	Categories
Education Level	Primary, Secondary, Graduate
Satisfaction Level	Poor, Average, Good, Excellent
Economic Status	Low, Middle, High
Class Rank	First, Second, Third

> **Important Point**
> 'Excellent' is better than 'Good', but the exact numerical difference between these categories is unknown.

1.5 Nominal vs. Ordinal Data

Basis	Nominal Data	Ordinal Data
Meaning	Categories without order	Categories with order
Ranking	Not possible	Possible
Comparison	Only equality	Equality and ranking
Examples	Gender, Blood Group	Grades, Satisfaction Levels

1.6 Binary (Dichotomous) Data

A special type of categorical data having only two possible categories is called binary or dichotomous data. It is a special case of nominal data.

Variable	Categories
Gender	Male, Female
Response	Yes, No
Result	Pass, Fail
Smoking Habit	Smoker, Non-Smoker

1.7 Coding of Categorical Data

For statistical analysis and data entry, categorical data are often converted into numerical codes. This practice is called coding. It is important to remember that these codes are labels only — they do not represent actual numerical quantities.

Category	Code
Male	1
Female	0
Poor (Satisfaction)	1
Average (Satisfaction)	2
Good (Satisfaction)	3
Excellent (Satisfaction)	4

> **Note**
> Numerical codes assigned to categories are only labels. Arithmetic operations on coded values (e.g., computing the mean of satisfaction codes) may not always be meaningful without justification.

1.8 Representation of Categorical Data

Categorical data are commonly represented using the following methods:

- Frequency Tables
- Bar Charts
- Pie Charts
- Contingency Tables

Example — Frequency Table:

Blood Group	Frequency
A	12
B	10
AB	5
O	18

1.9 Applications of Categorical Data

Categorical data are widely used across many fields:

Field	Example Application
Medical Research	Blood group analysis, disease classification
Marketing	Customer preference surveys
Education	Student grading and stream classification
Election Studies	Voting preference analysis
Business Analytics	Product category analysis

1.10 Advantages and Limitations

Advantages:

- Easy to collect and record
- Useful for classification and grouping
- Suitable for surveys and questionnaires
- Helpful in decision-making processes
- Simple to interpret for non-technical audiences

Limitations:

- Arithmetic operations are limited or non-meaningful
- Detailed numerical information is absent
- Range of applicable statistical techniques is comparatively limited
- Coding may sometimes oversimplify complex information

Part A — Summary

- Categorical data classify observations into groups based on qualitative characteristics.
- They are qualitative in nature; the values are labels, not measurements.
- Nominal data have no order; ordinal data have a meaningful order.
- Binary data are a special case with exactly two categories.
- Analysis primarily uses frequencies, proportions, and percentages.

PART B: Frequency Distributions and Graphical Representation

Course Outcomes (COs) Mapping

CO No.	Course Outcome
CO1	Understand frequency distribution of categorical data.
CO2	Construct one-way and two-way frequency tables for categorical variables.
CO3	Represent categorical data graphically using appropriate diagrams.
CO4	Interpret categorical data using tables and graphs.

1.11 Introduction to Frequency Distributions

In categorical data analysis, raw data must first be organized into a structured form before meaningful analysis can take place. The most fundamental organizational tool is the frequency distribution. Graphical representations then complement frequency distributions by providing visual insight into patterns, comparisons, and relationships.

> **Definition**
> A frequency distribution is a systematic arrangement of data showing the number of observations (the frequency) belonging to each category.

1.12 Types of Frequency Distributions

1.12.1 One-Way Frequency Distribution

A one-way frequency table shows the frequencies for a single categorical variable. It involves simple classification of observations into the categories of one variable.

Example — Distribution of students according to blood group:

Blood Group	Frequency
A	10
B	8
AB	4
O	12
Total	34

Characteristics:

- Involves only one categorical variable
- Simple classification and easy to construct
- Immediately shows which category is most or least frequent

1.12.2 Two-Way Frequency Distribution (Contingency Table)

A two-way frequency table, also called a contingency table, shows frequencies for two categorical variables simultaneously. It is the most important tool for studying relationships between categorical variables.

Example — Classification of students according to Gender and Result:

Gender	Pass	Fail	Total
Male	18	2	20
Female	15	5	20
Total	33	7	40

Characteristics:

- Involves two categorical variables
- Helps study the association between the two variables
- Each cell contains the joint frequency of the two categories
- Row and column totals are called marginal totals

1.12.3 Relative Frequency Distribution

Relative frequency expresses each category's frequency as a proportion or percentage of the total number of observations. This allows meaningful comparisons across groups of different sizes.

Formulae

Relative Frequency = Frequency ÷ Total Frequency Percentage Frequency = (Frequency ÷ Total Frequency) × 100

Example:

Blood Group	Frequency	Relative Frequency	Percentage (%)
A	10	10/34 = 0.2941	29.41%
B	8	8/34 = 0.2353	23.53%
AB	4	4/34 = 0.1176	11.76%
O	12	12/34 = 0.3529	35.29%
Total	34	1.0000	100%

1.13 Graphical Representation of Categorical Data

Graphical methods provide visual summaries of categorical data that make patterns and comparisons immediately apparent. The main graphical methods for categorical data are:

- Bar Diagram
- Pie Chart
- Segmented (Component) Bar Diagram
- Multiple Bar Diagram
- Pareto Chart

1.13.1 Bar Diagram

> **Definition**
> A bar diagram represents categorical data using rectangular bars of equal width. The height (or length) of each bar is proportional to the frequency of the corresponding category.

Characteristics:

- Bars are separated by equal spaces
- Most suitable for nominal and ordinal categorical data
- Allows easy comparison of frequencies across categories

Example:

Department	Number of Students
Computer Science	40
Information Technology	35
Data Science	30
AI & ML	25

Types of Bar Diagrams:

- Simple Bar Diagram — one bar per category
- Multiple Bar Diagram — two or more bars per category for comparison
- Component (Segmented) Bar Diagram — each bar is subdivided into parts

1.13.2 Pie Chart

Definition
A pie chart is a circular diagram divided into sectors, where the area of each sector is proportional to the frequency of the corresponding category.

Formula
Central Angle for a category = (Frequency ÷ Total Frequency) × 360°

Example — Calculating angles:

Category	Frequency	Central Angle
A	20	(20/100) × 360° = 72°
B	30	(30/100) × 360° = 108°
C	50	(50/100) × 360° = 180°
Total	100	360°

Characteristics:

- Represents proportions and percentages visually
- Best suited for a small number of categories (typically fewer than six)
- Useful when emphasizing each category's share of the whole

1.13.3 Segmented and Multiple Bar Diagrams

A segmented bar diagram represents subcategories within each category using different color segments stacked within the same bar. It is useful for displaying two-way classified data.

A multiple bar diagram places two or more bars side by side for each category, enabling direct comparison of subcategories. For example, comparing pass and fail rates across different classes.

1.13.4 Pareto Chart

A Pareto chart is a bar chart in which the categories are arranged in descending order of frequency, from the most frequent to the least frequent. A cumulative frequency line is often added.

Characteristics:

- Highlights the most important categories at a glance
- Widely used in quality control and business analysis
- Helps apply the 'vital few vs. trivial many' principle

1.14 Advantages and Limitations of Graphical Representation

Advantages:

- Provides easy and intuitive understanding of data
- Allows quick comparison between categories
- Attractive and suitable for reports and presentations
- Helps identify patterns, trends, and anomalies

Limitations:

- Exact values may not be directly readable from the graph
- Misleading scales or truncated axes can distort interpretation
- May oversimplify complex datasets
- Not suitable for highly detailed quantitative analysis

1.15 Interpreting Graphs

When interpreting graphical representations of categorical data, follow these guidelines:

- Compare the heights or sizes of bars/sectors carefully
- Identify the largest and smallest categories
- Look for patterns, clusters, or outliers
- Always check scales, labels, and titles
- Consider relative frequencies (proportions) alongside raw counts

Part B — Summary

- A frequency distribution systematically shows the number of observations in each category.
- One-way tables summarize one variable; contingency tables (two-way) summarize two variables simultaneously.
- Relative frequencies express proportions or percentages, enabling fair comparison across different group sizes.
- Bar diagrams and pie charts are the primary graphical tools for categorical data.
- Pareto charts highlight the most significant categories by ordering bars by descending frequency.

PART C: Cross-Classification, Binomial and Multinomial Distributions

Course Outcomes (COs) Mapping

CO No.	Course Outcome
CO1	Understand cross-classification of categorical variables.
CO2	Construct and interpret contingency tables.
CO3	Understand the concept and properties of binomial distribution.
CO4	Explain multinomial distribution and its applications in categorical data analysis.

1.16 Cross-Classification of Categorical Variables

Definition
Cross-classification is the arrangement of observations according to two or more categorical variables simultaneously. The resulting table is called a contingency table.

Cross-classification helps to:

- Study relationships and associations between categorical variables
- Compare distributions across groups
- Summarize large datasets effectively
- Provide the foundation for chi-square tests of independence

1.17 Types of Cross-Classification

1.17.1 Two-Way Classification

Data are classified according to two categorical variables simultaneously.

Example — Classification by Gender and Stream:

Gender	Science	Commerce	Arts
Male	20	15	10
Female	18	12	8

1.17.2 Three-Way and Multi-Way Classification

Three-way classification involves three categorical variables (e.g., Gender, Stream, and Residence Area). Multi-way classification involves more than three variables and is used in advanced categorical data analysis.

1.18 Components of a Contingency Table

A contingency table has the following components:

Component	Description	Example (from table below)
Cell Frequency	Number of observations in a specific row-column combination	18 (Male & Pass)
Row Total (Marginal)	Total for each row	20 (all Male students)
Column Total (Marginal)	Total for each column	33 (all Pass students)
Grand Total	Total of all observations	40 (all students)

Example Contingency Table:

Gender	Pass	Fail	Total
Male	18	2	20
Female	15	5	20
Total	33	7	40

1.19 Binomial Distribution

The binomial distribution is a fundamental discrete probability distribution for categorical data where each trial has exactly two possible outcomes (binary categorical data).

Definition

A random variable X follows a binomial distribution if: (1) the number of trials n is fixed, (2) all trials are independent, (3) each trial has exactly two outcomes (success or failure), and (4) the probability of success p remains constant across trials.

Notation: $X \sim B(n, p)$

where n = number of trials, p = probability of success, $q = 1 - p$ = probability of failure.

Probability Mass Function (PMF)

$P(X = x) = C(n, x) \times p^x \times q^{(n-x)}$, for $x = 0, 1, 2, ..., n$ where $C(n, x) = n! / [x! (n-x)!]$ is the binomial coefficient.

Mean and Variance:

Parameter	Formula
Mean	$\mu = np$
Variance	$\sigma^2 = npq$
Standard Deviation	$\sigma = \sqrt{(npq)}$

Worked Example

Problem: A fair coin is tossed 5 times. Find the probability of obtaining exactly 3 heads.

Given: $n = 5$, $p = 0.5$ (probability of head), $q = 0.5$, $x = 3$

$P(X = 3) = C(5, 3) \times (0.5)^3 \times (0.5)^2$

$= 10 \times 0.125 \times 0.25$

$= 0.3125$

Applications of Binomial Distribution:

- Quality control (defective vs. non-defective items)
- Medical diagnosis (positive vs. negative test results)
- Survey analysis (agree vs. disagree responses)

- Election studies (vote vs. no-vote)

1.20 Multinomial Distribution

The multinomial distribution is a natural extension of the binomial distribution to situations where each trial can result in more than two possible outcomes.

Definition

A random variable follows a multinomial distribution if: (1) the number of trials n is fixed, (2) all trials are independent, (3) each trial can result in one of k > 2 possible outcomes, and (4) the probability of each outcome remains constant.

Notation:

If X_1 , X_2 , ..., X_k denote the number of times outcomes 1, 2, ..., k occur in n trials, then (X_1 , X_2 , ..., X_k) follows a multinomial distribution with parameters n, p_1 , p_2 , ..., p_k where p_1 + p_2 + ... + p_k = 1.

Probability Mass Function

$P(X_1 = x_1 , X_2 = x_2 , ..., X_k = x_k) = [n! / (x_1 ! x_2 ! \cdots x_k !)] \times p_1 ^x_1 \times p_2 ^x_2 \times \cdots \times p_k ^x_k$
where $x_1 + x_2 + ... + x_k = n$

Examples of situations modelled by the multinomial distribution:

- Rolling a fair die (6 outcomes)
- Blood group classification (4 outcomes: A, B, AB, O)
- Voting preference among multiple political parties
- Product preference surveys with more than two options

1.21 Binomial vs. Multinomial Distribution

Basis	Binomial Distribution	Multinomial Distribution
Number of Outcomes	Exactly two (binary)	More than two (k ≥ 2)
Example	Pass / Fail; Yes / No	Blood groups A, B, AB, O
Parameters	n, p (and q = 1 − p)	n, p_1 , p_2 , ..., p_k
Application	Binary categorical data	Multi-category categorical data
Special Case	Special case of multinomial (k = 2)	Generalizes binomial to k categories

1.22 Applications in Categorical Data Analysis

Binomial Distribution is used for:

- Modeling binary categorical variables (two-category outcomes)
- Success/failure experiments in quality control
- Serving as the basis for logistic regression (binary response)

Multinomial Distribution is used for:

- Modeling multi-category outcomes (k > 2 categories)
- Analyzing contingency tables under specific hypotheses
- Serving as the basis for multinomial logistic regression
- Modeling word frequencies in text classification

Part C — Summary

- Cross-classification organizes data according to two or more categorical variables simultaneously.
- Contingency tables display joint and marginal frequencies and form the basis for tests of association.
- The binomial distribution models experiments with exactly two outcomes; its parameters are n (trials) and p (probability of success).
- The multinomial distribution generalizes the binomial to k > 2 outcome categories.
- Both distributions are foundational probability models for categorical data analysis and inference.

Chapter Summary

Topic	Key Takeaway
Categorical Data	Qualitative data classifying observations into distinct groups or categories
Nominal Data	Categories with no natural order (e.g., blood group, nationality)
Ordinal Data	Categories with a meaningful order (e.g., satisfaction level, grades)
Binary Data	Special case with exactly two categories (e.g., pass/fail, yes/no)
Frequency Distribution	Systematic table showing counts for each category
Contingency Table	Two-way table for cross-classifying two categorical variables

Relative Frequency	Category frequency expressed as a proportion of total observations
Bar Diagram	Bars of height proportional to category frequency
Pie Chart	Circular diagram with sectors proportional to category proportions
Pareto Chart	Bar chart with categories in descending frequency order
Binomial Distribution	Probability model for n independent trials each with two outcomes
Multinomial Distribution	Extension of binomial to trials with more than two outcomes

This chapter has laid the conceptual and statistical foundations for categorical data analysis. The concepts of nominal and ordinal measurement, frequency distributions, contingency tables, and the binomial and multinomial probability distributions will be revisited and built upon in subsequent chapters as we develop more advanced methods for analyzing, testing, and modelling categorical data.